

CSC 220 Assignment 02 In Class Starting Checkpoint

Tuesday September 29 2026 • 40 minutes • 16 points

Begin Assignment 02 in class. By the end of the work period, demonstrate a working start on your own generic doubly linked list, show test results, and submit a link to your shared project files. This checkpoint is graded separately from the 100 point assignment.

Work and submission

Work individually or with one Assignment 02 partner. Create either a shared Google Drive folder or a GitHub repository for your source code. Both partners must be able to access the current files. Submit one link through the in class task in Canvas, with both students' names. Show your program to the instructor before leaving; each student must explain one pointer update.

Required work during the 40 minutes

1. Create `DoublyLinkedList<T>` with a private node containing data, next, and previous references. Include head, tail, and size fields.
2. Implement `addFirst(T value)` and `addLast(T value)`. Handle both an empty list and a list that already has nodes.
3. Write and run a small test using `DoublyLinkedList<Integer>`. Add an item to an empty list, another at the front, and another at the back.
4. Show the expected and actual order and size after the three additions. You may write a temporary display method to make the result visible. Be ready to explain how the links and endpoints changed.

For example, after `addFirst(10)`, `addFirst(5)`, and `addLast(20)`, the expected order is 5, 10, 20 and the expected size is 3. Code that compiles alone does not demonstrate that it works: run the test and compare the actual result with your expectation.

Grading

Evidence demonstrated	Points
Generic node and list structure, including head, tail, and size	2
<code>addFirst</code> handles empty and nonempty lists	2
<code>addLast</code> handles empty and nonempty lists	2
Runnable test shows the three additions, resulting order and size, with expected versus actual results	5
Each student explains a pointer update	2
Shared Google Drive folder or GitHub repository link submitted and accessible to both partners	3
Total	16

Assignment 02 deadline

The full 100 point Assignment 02 is due Monday October 12 2026 at 11:59 pm. This deadline allows time to prepare for Thursday's term exam and accounts for the Friday October 9 fall holiday. The in class checkpoint is due at the end of today's work period. Partial credit is available for demonstrated progress.